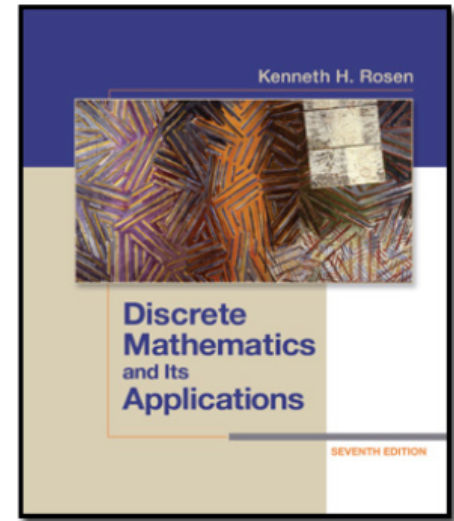


Chapter 1, Part III (Proofs)-Sec 1.6

Rules of Inference



Chapter Summary

- Propositional Logic
- Predicate Logic
- Proofs

Revisiting the Socrates Example

- Two premises:
 - “All men are mortal.”
 - “Socrates is a man.”
- A conclusion:
 - “Socrates is mortal.”
- How do we get the conclusion from the premises?

The Argument

A valid argument

$$\forall x (Man(x) \rightarrow Mortal(x))$$

• *premises*

$$Man(Socrates)$$

$$\therefore Mortal(Socrates)$$

• *conclusion*

Valid Arguments

The *rules of inference* are the essential building blocks in the construction of valid arguments.

1. Propositional Logic

Inference Rules

2. Predicate Logic

Inference rules for propositional logic PLUS
inference rules to handle *variables* and *quantifiers*.

Arguments in Propositional Logic

- An *argument*: a sequence of propositions.
 - All but the final proposition are called *premises*.
 - The last statement is the *conclusion*.
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Inference rules are all *simple argument forms* that will be used to construct more complex argument forms.

Rules of Inference for Propositional Logic:

Modus Ponens

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Corresponding Tautology:
 $(p \wedge (p \rightarrow q)) \rightarrow q$

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Example:

Let p be “It is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“It is snowing.”

“Therefore, I will study discrete math.”

Modus Tollens

$$\begin{array}{c} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

Corresponding Tautology:
 $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$

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Example:

Let p be “it is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“I will not study discrete math.”

“Therefore , it is not snowing.”

Hypothetical Syllogism

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Corresponding Tautology:
 $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

Hypothetical Syllogism

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Corresponding Tautology:
 $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

Example:

Let p be "it snows."

Let q be "I will study discrete math."

Let r be "I will get an A."

"If it snows, then I will study discrete math."

"If I study discrete math, I will get an A."

"Therefore, If it snows, I will get an A."

Disjunctive Syllogism

$$p \vee q$$

$$\neg p$$

$$\therefore q$$

Corresponding Tautology:

$$(\neg p \wedge (p \vee q)) \rightarrow q$$

Disjunctive Syllogism

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$$\neg p$$

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Example:

Let p be "I will study discrete math."

Let q be "I will study English literature."

"I will study discrete math or I will study English literature."

"I will not study discrete math."

"Therefore, I will study English literature."

Addition

$$\frac{p}{\therefore p \vee q}$$

Corresponding Tautology:
 $p \rightarrow (p \vee q)$

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$$\frac{p}{\therefore p \vee q}$$

Corresponding Tautology:

$$p \rightarrow (p \vee q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will visit Las Vegas.”

“I will study discrete math.”

“Therefore, I will study discrete math or I will visit Las Vegas.”

Simplification

$$\frac{p \wedge q}{\therefore q}$$

Corresponding Tautology:

$$(p \wedge q) \rightarrow p$$

Simplification

$$\frac{p \wedge q}{\therefore q}$$

Corresponding Tautology:
 $(p \wedge q) \rightarrow p$

Example:

Let p be "I will study discrete math."

Let q be "I will study English literature."

"I will study discrete math and English literature"

"Therefore, I will study discrete math."

Conjunction

$$\frac{p}{q} \quad \frac{q}{\therefore p \wedge q}$$

Corresponding Tautology:
 $((p) \wedge (q)) \rightarrow (p \wedge q)$

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Corresponding Tautology:
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Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math.”

“I will study English literature.”

“Therefore, I will study discrete math and I will study English literature.”

Resolution

$$\frac{\neg p \vee r \quad p \vee q}{\therefore q \vee r}$$

Corresponding Tautology:
 $((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$

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$$\frac{\neg p \vee r \quad p \vee q}{\therefore q \vee r}$$

Corresponding Tautology:
 $((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$

Example:

Let p be “I will study discrete math.”

Let r be “I will study English literature.”

Let q be “I will study databases.”

“I will not study discrete math or I will study English literature.”

“I will study discrete math or I will study databases.”

“Therefore, I will study databases or I will study English literature.”

Valid Arguments

Example 1: From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that q is a conclusion.

Solution:

Step	Reason
1. $p \wedge (p \rightarrow q)$	Premise
2. p	Simplification using (1)
3. $p \rightarrow q$	Simplification using (1)
4. q	Modus Ponens using (2) and (3)

Valid Arguments

Example 2:

- With these hypotheses:
 - “It is not sunny this afternoon and it is colder than yesterday.”
 - “We will go swimming only if it is sunny.”
 - “If we do not go swimming, then we will take a canoe trip.”
 - “If we take a canoe trip, then we will be home by sunset.”
- Using the inference rules, construct a valid argument for the conclusion:
 - “We will be home by sunset.”

Valid Arguments

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- Using the inference rules, construct a valid argument for the conclusion:
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Solution:

1. Choose propositional variables:
 - p : “It is sunny this afternoon.” r : “We will go swimming.”
 - t : “We will be home by sunset.”
 - q : “It is colder than yesterday.” s : “We will take a canoe trip.”
2. Translation into propositional logic:

Hypothesis: ?

Conclusion: ?

Continued on next slide →

Valid Arguments

Example 2:

- With these hypotheses:
 - “It is not sunny this afternoon and it is colder than yesterday.”
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Solution:

1. Choose propositional variables:
 - p : “It is sunny this afternoon.” r : “We will go swimming.”
 - t : “We will be home by sunset.”
 - q : “It is colder than yesterday.” s : “We will take a canoe trip.”
2. Translation into propositional logic:

Hypotheses: $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$
Conclusion: t

Continued on next slide →

Valid Arguments

Hypotheses: $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, $s \rightarrow t$
Conclusion: t

3. Construct the Valid Argument

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Quantified Statements

Universal Instantiation (UI)

$$\frac{\forall x P(x)}{\therefore P(c)}$$

Quantified Statements

Universal Instantiation (UI)

$$\frac{\forall x P(x)}{\therefore P(c)}$$

Example:

Our domain consists of all dogs and Fido is a dog.

“All dogs are cuddly.”

“Therefore, Fido is cuddly.”

Quantified Statements

Universal Generalization (UG)

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$$

Used often implicitly in Mathematical Proofs.

Quantified Statements

Existential Instantiation (EI)

$$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$$

Quantified Statements

Existential Instantiation (EI)

$$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$$

Example:

“There is someone who got an A in the course.”

“Let’s call her a and say that a got an A”

Existential Generalization (EG)

$$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$$

Existential Generalization (EG)

$$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$$

Example:

“Michelle got an A in the class.”

“Therefore, someone got an A in the class.”

Quantified Statements

Example 2: Use the rules of inference to construct a valid argument showing that the conclusion

“Someone who passed the first exam has not read the book.”

follows from the premises

“A student in this class has not read the book.”

“Everyone in this class passed the first exam.”

Continued on next slide →

Using Rules of Inference

Solution: Let

$C(x)$: “ x is in this class,”

$B(x)$: “ x has read the book,”

$P(x)$: “ x passed the first exam.”

“Someone who passed the first exam has not read the book.”

follows from the premises

“A student in this class has not read the book.”

“Everyone in this class passed the first exam.”

$$\frac{\begin{array}{l} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x(P(x) \wedge \neg B(x))}$$

Continued on next slide →

Using Rules of Inference

Valid Argument:

Step	Reason
1. $\exists x(C(x) \wedge \neg B(x))$	Premise
2. $C(a) \wedge \neg B(a)$	EI from (1)
3. $C(a)$	Simplification from (2)
4. $\forall x(C(x) \rightarrow P(x))$	Premise
5. $C(a) \rightarrow P(a)$	UI from (4)
6. $P(a)$	MP from (3) and (5)
7. $\neg B(a)$	Simplification from (2)
8. $P(a) \wedge \neg B(a)$	Conj from (6) and (7)
9. $\exists x(P(x) \wedge \neg B(x))$	EG from (8)

Returning to the Socrates Example

$$\forall x (Man(x) \rightarrow Mortal(x))$$

$$Man(Socrates)$$

$$\therefore Mortal(Socrates)$$

Solution for Socrates Example

Valid Argument

Step

1. $\forall x(Man(x) \rightarrow Mortal(x))$
2. $Man(Socrates) \rightarrow Mortal(Socrates)$
3. $Man(Socrates)$
4. $Mortal(Socrates)$

Reason

Premise
UI from (1)
Premise
MP from (2)
and (3)

Universal Modus Ponens

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ P(a), \text{ where } a \text{ is a particular} \\ \text{element in the domain} \\ \hline \therefore Q(a) \end{array}$$

Universal Modus Ponens combines universal instantiation and modus ponens into one rule.

This rule could be used in the Socrates example.